Phase Transition in Nonparametric Minimax Rates for Covariate Shifts on Approximate Manifolds

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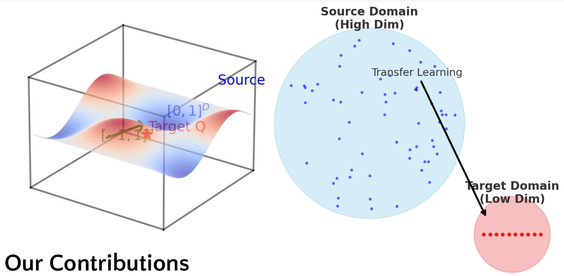
Motivation

Transfer Learning: Leverage knowledge from **source** (abundant) to improve performance on **target** (scarce).

Motivations:

- **Structured covariate shift**: Target data are scarce but concentrated near a low-dimensional manifold, while source data are abundant in a higher-dim space.
- **No density ratios**: Different intrinsic dimensions make dQ/dP undefined, so standard transfer methods fail.
- **Two levers to exploit**: Combine source with manifold + smoothness structure to beat target-only rates.
- **Approximate manifolds**: Targets lie *near* (not exactly on) a manifold; need to account for distance ρ_n.
- **Rate-level understanding**: Seek sharp minimax rates to characterize when transfer helps, in terms of $(n_P, n_Q, \beta, d, D, \rho_n)$



Our Contributions

- **Phase transition in minimax rates**: Discover a sharp transition governed by ρ_n and an effective scale κ_n^*, separating two regimes of transfer benefit.
- **Optimal rates with approximate manifolds**: Establish minimax rates that interpolate between intrinsic- and ambient-dimensional behaviors
- **Rate-optimal estimator**: Develop an estimator that achieves the optimal rate, improving over prior estimators when $\beta > 1$.
- **Adaptive procedure**: Design a data-driven estimator, attaining near-optimal rates without knowing (β, d).

Local Polynomial Regression

Our Proposed Estimator (Modified Version of LPR):

Problem: Target may lie on a **low-dimensional manifold** inside source; Standard LPR → unstable, ill-conditioned

Solution: Truncated Estimator

$$\hat{f}(x^*) := \begin{cases} e_1^\top (Z^\top W Z)^{-1} Z^\top W Y & \text{if } \lambda_{min}(\boldsymbol{Z}^\top \boldsymbol{W} \boldsymbol{Z}) \geq n\tau_n\psi_n \\ 0 & \text{otherwise} \end{cases}$$

with bandwidth h, design matrix Z, weight matrix W

✓ $\psi_n := \frac{n_P}{n} + \frac{n_Q}{n}(\rho_n \vee h_n)^{d-D}$ is a normalizing constant

✓ $\tau_n := \left(n_P h_n^D + n_Q(\rho_n \vee h_n)^{d-D} h_n^D\right)^{-p}$ for some $p > 1$

Assumptions

Goal: Estimate $\boldsymbol{\beta}$**-Hölder** $\boldsymbol{f^*(x) = E[Y \mid X = x]}$

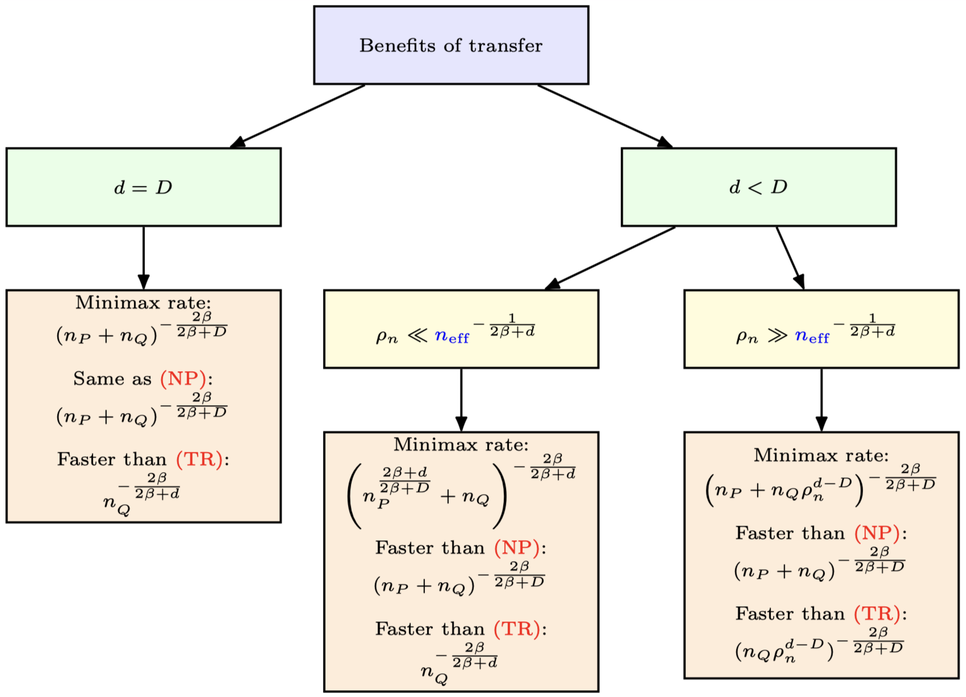
- Covariate Shift: $\boldsymbol{P_X \neq Q_X}$ but
$$\boldsymbol{E_P[Y|X] = E_Q[Y|X] = f^*(X)}$$
- Source distribution $\boldsymbol{P_X}$: Supported on $[-2,2]^D$ **with** n_P samples
- Target distribution $\boldsymbol{Q_X}$: Supported on approx. manifold $\boldsymbol{\mathcal{M}_{\rho_n}}$ **with** n_Q samples, $n_P \gg n_Q$
  - ✓ Model: $\boldsymbol{X = \phi(V) + \rho_n U}$ where smooth embedding $\boldsymbol{\phi}: [-1,1]^d \to [-1,1]^D$
  - ✓ Jacobian well-conditioned:
$$\boldsymbol{c^{-1} \leq \lambda_{min}(\nabla\phi^\top \nabla\phi) \leq \lambda_{max}(\cdot) \leq c}$$
- Moment Condition: For both source & target, fourth conditional moment bounded.

Main Results

Minimax Rates: We prove that the estimator achieves the minimax-optimal rate. Specifically, let $\kappa_n^* = \left(n_P^{(2\beta+d)/(2\beta+D)} + n_Q\right)^{-1/(2\beta+d)}$, under standard assumptions, we have:

$$\inf_{\hat{f}} \sup E\left(\hat{f}(x^*) - f(x^*)\right)^2 \asymp \begin{cases} \left(n_P + n_Q \rho_n^{d-D}\right)^{-2\beta/(2\beta+D)} & if\ \rho_n \geq C\kappa_n^* \\ \left(n_P^{(2\beta+d)/(2\beta+D)} + n_Q\right)^{-2\beta/(2\beta+d)} & if\ \rho_n \leq C\kappa_n^* \end{cases}$$

- **high-noise** (ρ_n large): rates scale with ambient dim (D) ⇒ **better than pooled & target-only**
- **low-noise** (ρ_n small): rates switch to **intrinsic dim (d)** ⇒ **faster than target-only**



Adaptive Algorithm

Adaptive Algorithm: → handles unknown smoothness + manifold dim

Input: Source$\{(X_i, Y_i)\}_{i=1}^{n_P}$; Target $\{(X_i, Y_i)\}_{i=n_P+1}^{n_P+n_Q}$; Parameters: $\boldsymbol{D, C_h, C_1, \widetilde{\mathcal{B}}, k}$

Step 1 : Intrinsic Dimension:

Local estimate via k-NN:$\hat{d}(X_i) = \left[\log 2 / \log\left(\frac{r^{(k)}(X_i)}{r^{(\lfloor k/2 \rfloor)}(X_i)}\right)\right]$;

Global estimate: $\hat{d} = \left[\frac{1}{n_Q}\sum_i \left(\hat{d}(X_i) \wedge D\right)\right]$

Step 2: Candidate Estimators: For each $\boldsymbol{\beta} \in \widetilde{\mathcal{B}}$:

Bandwidth:$h_{n,\tilde{\beta},\hat{d}} = C_h\left(\left(n_Q + n_P^{(2\tilde{\beta}+\hat{d})/(2\tilde{\beta}+D)}\right)/\log n\right)^{-1/(2\tilde{\beta}+\hat{d})}$.

Step 3: Lepski's Method:

smoothness: $\hat{\beta} = \max\left\{\tilde{\beta} \in \widetilde{\mathcal{B}}: \left|\widehat{f_{h_{n,\tilde{\beta},\hat{d}}}}(x^*) - \widehat{f_{h_{n,\eta,\hat{d}}}}(x^*)\right| \leq C_1 h_{n,\eta,\hat{d}}^{\eta},\ \forall \eta \leq \tilde{\beta}\right\}$.

Step 4: Final Output

Adaptive bandwidth: $\widehat{h_n} = C_h\left(\left(n_Q + n_P^{(2\hat{\beta}+\hat{d})/(2\hat{\beta}+D)}\right)/\log n\right)^{-1/(2\hat{\beta}+\hat{d})}$;

Final estimator:$\widehat{f_{\text{adp}}}(x^*) := \widehat{f_{\widehat{h_n}}}(x^*)$

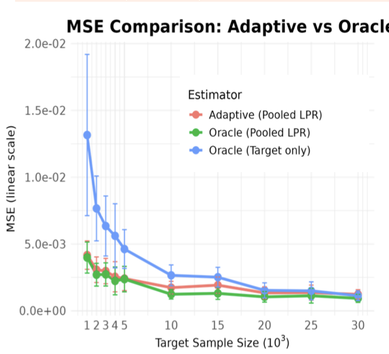
Experiment Results

Data Generation

- Source covariates: $X_i^{(P)} \sim \text{Unif}([0,1]^5)$, $i = 1, \ldots, n_P, n_P = 5000, \text{D} = 5$.
- Target covariates: sample $(Z_{1i}, Z_{2i}) \sim \text{Unif}([-1,1]^2)$, d = 2; set $X_i^{(Q)} = \phi(Z_{1i}, Z_{2i})$, $\phi(z_1, z_2) = (\frac{z_1+1}{2}, \frac{z_2+1}{2}, z_1^2, z_2^2, \frac{(z_1+1)(z_2+1)}{4})$, vary $n_Q \in \{1000, 2000, \ldots, 30000\}$.
- Response: $Y = f^*(X) + \epsilon$ where $\epsilon \sim N(0,1)$,
$$f^*(x) = \sum_{i=1}^{D} |x_i - x_i^*|^\beta, \beta = 2.5$$

Baselines (with true $\boldsymbol{\beta, d}$)

- Oracle (Pooled): uses source + target & Oracle (Target-only): only target



Key Findings:

- **Adaptive (Pooled) ≈ Oracle (Pooled)** Shows accuracy of adaptive method
- **Adaptive (Pooled) > Oracle (Target-only):** benefit of transfer learning; **Adaptive** outperforms **Target-only.**
- **Proposed (Combined)** consistently outperforms **Target-only**
- Gap is **largest when** n_P is **big and** n_Q **is small** → strong transfer gain
- Across all settings, adding source data never hurts — transfer helps